(I)(僅翻譯成中文而無解釋，得 0 分)

1. Cones vs Rods on the retina：

Cones 為視錐細胞，分布在 retina(視網膜)中央，主要與光視覺有關(用以辨識色彩，偵測物體細節)。數量較 Rods 少。

Rods 為視桿細胞，分布在 retina(視網膜)周邊，主要與暗視覺有關(光線明暗變化，物體型態變化或移動)。數量較 Cones 多。

2. Scotopic Vision：

暗視覺。在暗處(極低照度)下，人眼僅能依靠視桿細胞分辨光度灰階明暗的差異，無法分辨色彩。

3. Brightness Adaptation：

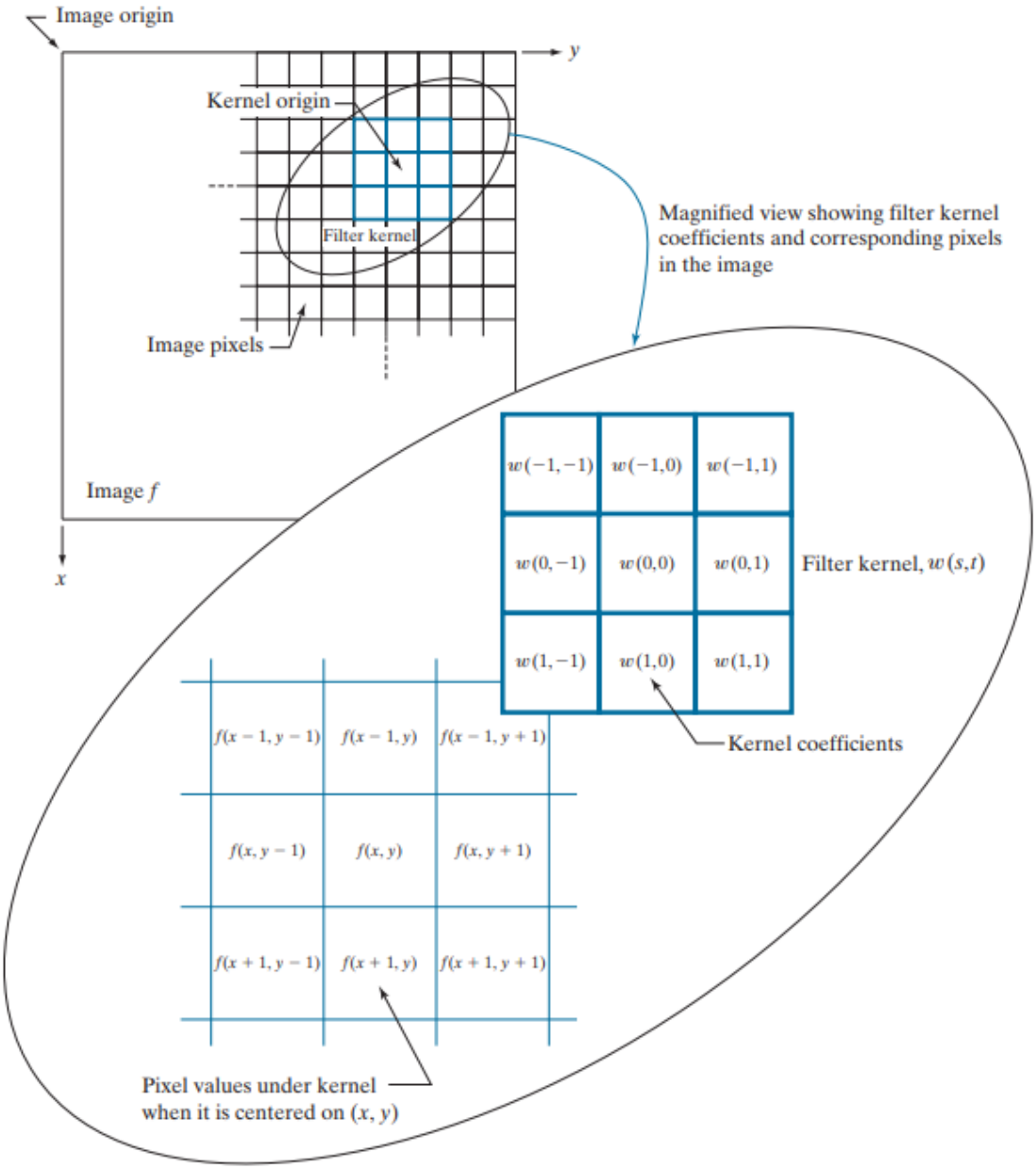
人的視覺系統會依照環境光度來改變感知光線的敏感度。

4. Spatial Correlation vs Convolution：

Correlation consists of moving the center of a kernel over an image, and computing the sum of products at each location.

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)$$

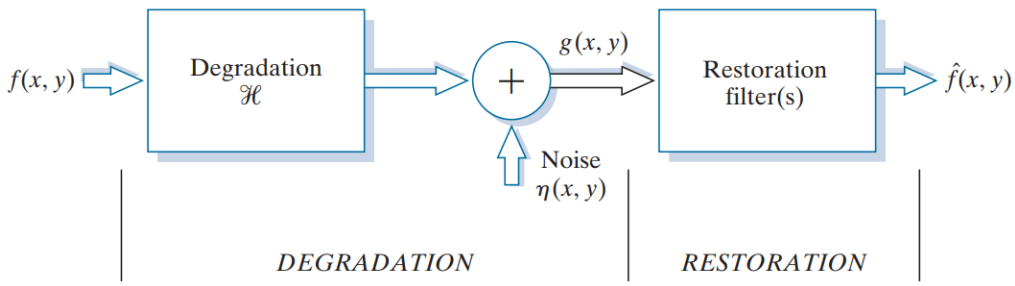
The mechanics of spatial convolution are the same, except that the correlation kernel is rotated by 180°. Thus, when the values of a kernel are symmetric about its center, correlation and convolution yield the same result.



5. Image Degradation vs. Restoration process：

Image Degradation:影像經過攝影機、掃描器或顯示器成像過程中，產生影像品質降低的現象。影像衰減的主要因素常為感測器與感光片的非線性、光學上的像差、影像移動、幾何畸變等所致。

Restoration process 為復原該品質降低影像之過程，其關係如下圖。



6. Salt-and-Pepper Noise：

隨機出現之黑點或白點的雜訊，可藉中值濾波器去除。

7. Geometric Mean Filter：

用來讓圖像平滑並且減少雜訊

$$G(x,y) = \left[\prod_{i,j\varepsilon S} S(i,j)\right]^{\frac{1}{mn}}$$

8. Gaussian Filter Kernel：

用來讓圖像模糊並去除高斯雜訊

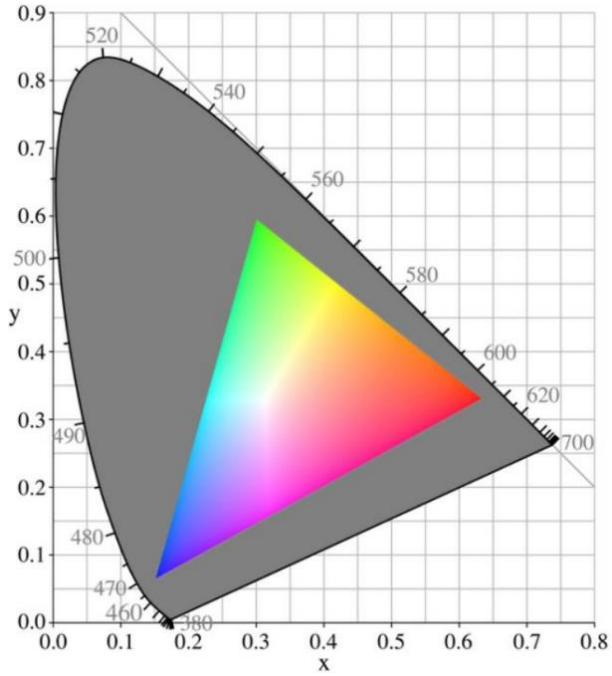
$$w(s,t) = G(s,t) = Ke^{-\frac{s^2+t^2}{2\sigma^2}}$$

9. Pseudocolor Image Processing：

因為人眼沒有辦法很好的辨識灰階值的程度，藉由在灰階影像上分配假色可以使該影像更好地詮釋其欲提供給人的訊息。

10. Color Gamut：

色域，即彩色顯示器可以顯示顏色的範圍。



11. Orthogonal Basis：

可展開(span)內積空間(inner product space)之基底(basis)，其內之各向量(vector)兩兩正交(orthogonal)，即內積為零。

12. Spatial and Temporal Redundancy：

Temporal Redundancy：在影片資料中，相鄰的 frame 與 frame 之間通常有很強的關連性，如代表背景的 pixel 可能在連續之 frame 皆相同，僅前景正在動之人物的 pixel 會變化，不同 frame 之背景 pixel 即有 redundancy。

Spatial Redundancy：在同一張 frame 之中，非人物邊界等之相鄰的 pixel 之間通常有很強的關連性，在此 frame 內該類 pixel 之間即有 redundancy。

13. Mathematical Morphology：

數學形態學，其可以提取 image component，例如邊界或骨架。或是描述影像中的區域形狀。基本運算如：erosion, dilation 等。

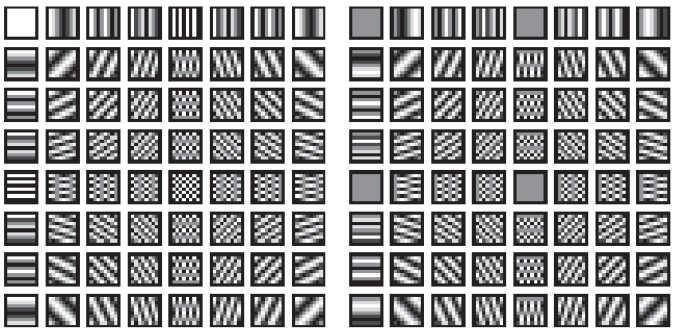
14. Watermarking：

將特定的可見或不可見之資訊嵌入影像中。浮水印可用來加入版權認證、鑑定該影像真偽(是否被改動過)，或是監控該影像在何時何地被播映等。

15. Basis Image：

將 DFT 之 N*N 轉換矩陣依照其線性組合來擺布成人眼較好觀察頻域的 N*N 個圖像。

$$\frac{1}{\sqrt{8}}\begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & -j & -j\omega & -1 & -\omega & j & j\omega \\ 1 & -j & -1 & j & 1 & -j & -1 & j \\ 1 & -j\omega & j & \omega & -1 & j\omega & -j & -\omega \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\omega & -j & j\omega & -1 & \omega & j & -j\omega \\ 1 & j & -1 & -j & 1 & j & -1 & -j \\ 1 & j\omega & j & -\omega & -1 & -j\omega & -j & \omega \end{bmatrix}$$



a b c

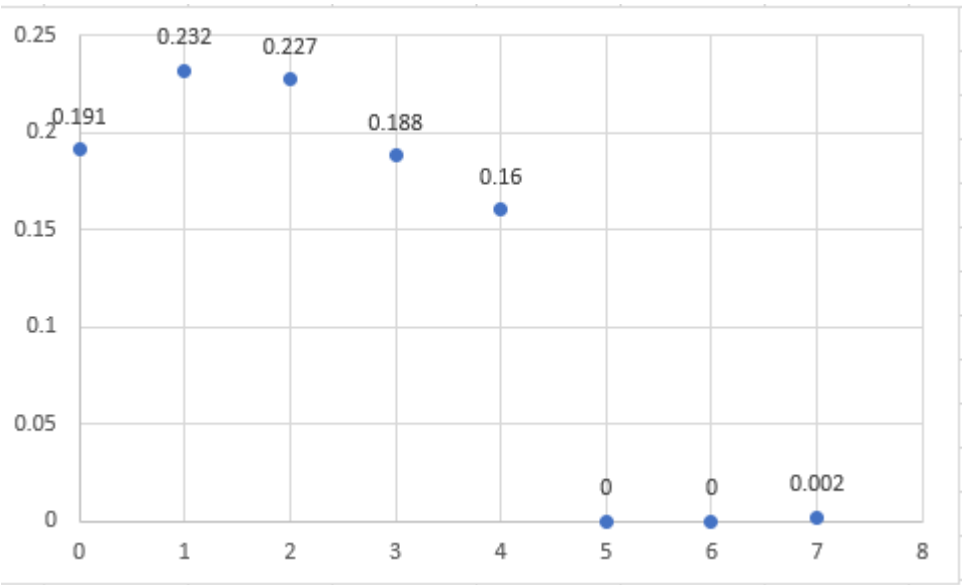
FIGURE 7.7 (a) Tranformation matrix $\mathbf{A}_F$ of the discrete Fourier transform for $N = 8$, where $\omega = e^{-j2\pi/8}$ or $(1-j)/\sqrt{2}$. (b) and (c) The real and imaginary parts of the DFT basis images of size 8×8. For clarity, a black border has been added around each basis image. For 1-D transforms, matrix $\mathbf{A}_F$ is used in conjunction with Eqs. (7-43) and (7-44); for 2-D transforms, it is used with Eqs. (7-41) and (7-42).

(II) 3-bit(L=8) Image, Total Pixels = 64x64 =4096

rk	n_k	$P_k(r_k)$	s_k	$\overline{s_k}$
0	781	$\frac{781}{4096} = 0.191$	1.337	1
1	951	$\frac{951}{4096} = 0.232$	2.961	3
2	930	$\frac{930}{4096} = 0.227$	4.55	5
3	768	$\frac{768}{4096}$ =0.188	5.866	6
4	656	$\frac{656}{4096} = 0.160$	6.986	7
5	0	$\frac{0}{4096} = 0$	6.986	7
6	0	$\frac{0}{4096} = 0$	6.986	7
7	10	$\frac{10}{4096} = 0.002$	7	7

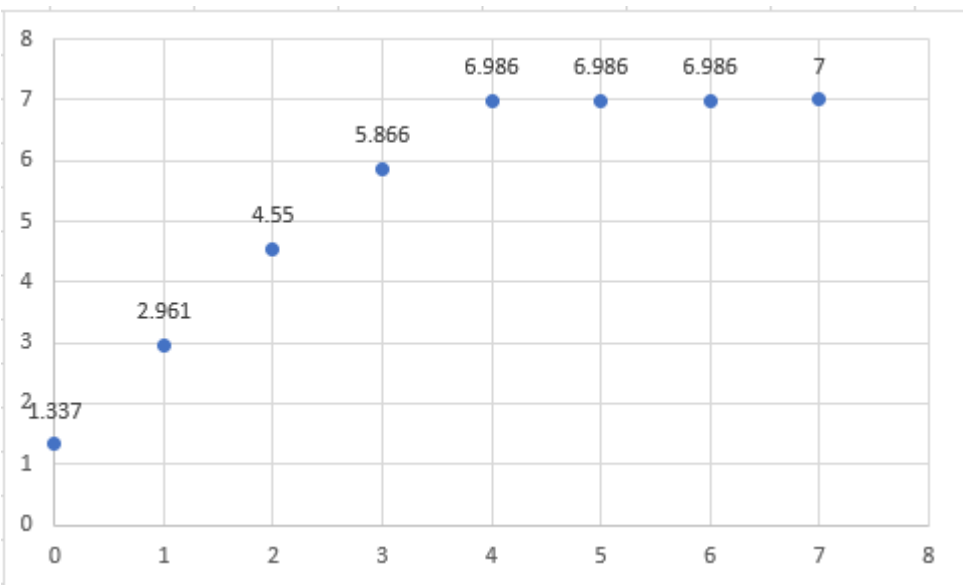
(1)PDF

r_k	n_k	$P_k(r_k)$
0	781	0.191
1	951	0.232
2	930	0.227
3	768	0.188
4	656	0.160
5	0	0
6	0	0
7	10	0.002



(2)SPV

s_0	1.337
s_1	2.961
s_2	4.55
s_3	5.866
s_4	6.986
s_5	6.986
s_6	6.986
s_7	7

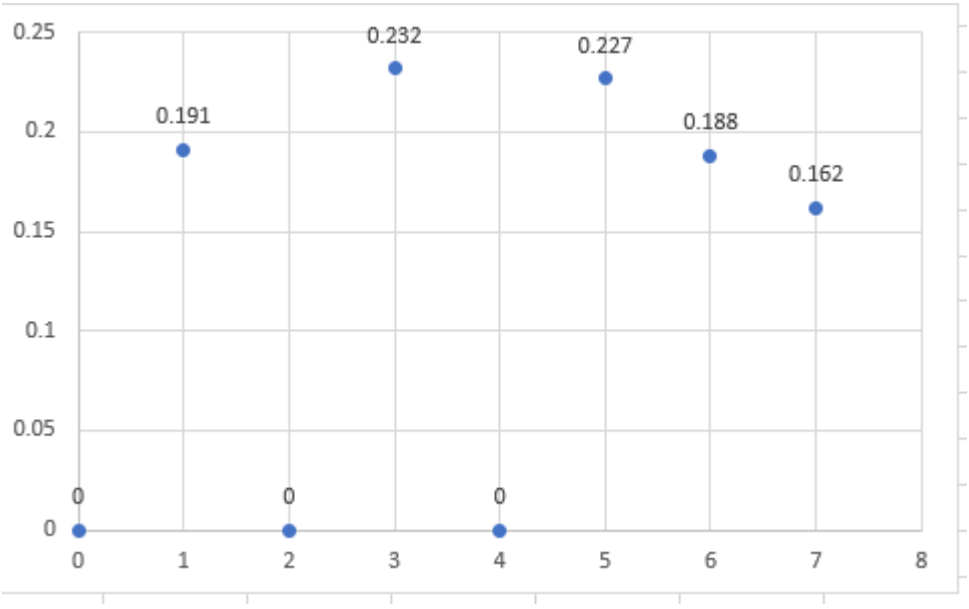


(3)

r_k	s_k
0	1
1	3
2	5
3	6
4	7
5	7
6	7
7	7

(4)

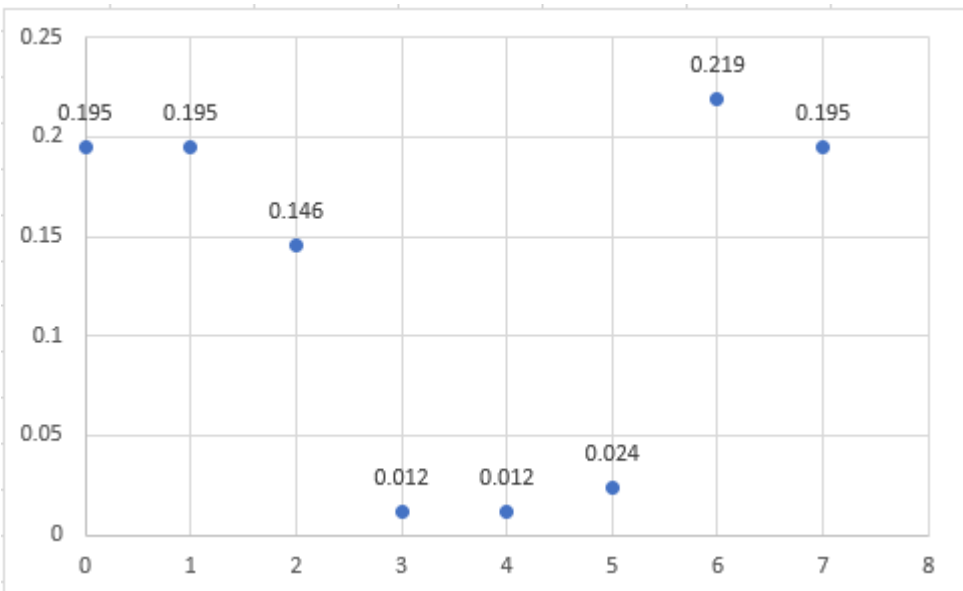
s_k	$P_s(s_k)$
0	0
1	0.191
2	0
3	0.232
4	0
5	0.227
6	0.188
7	0.162



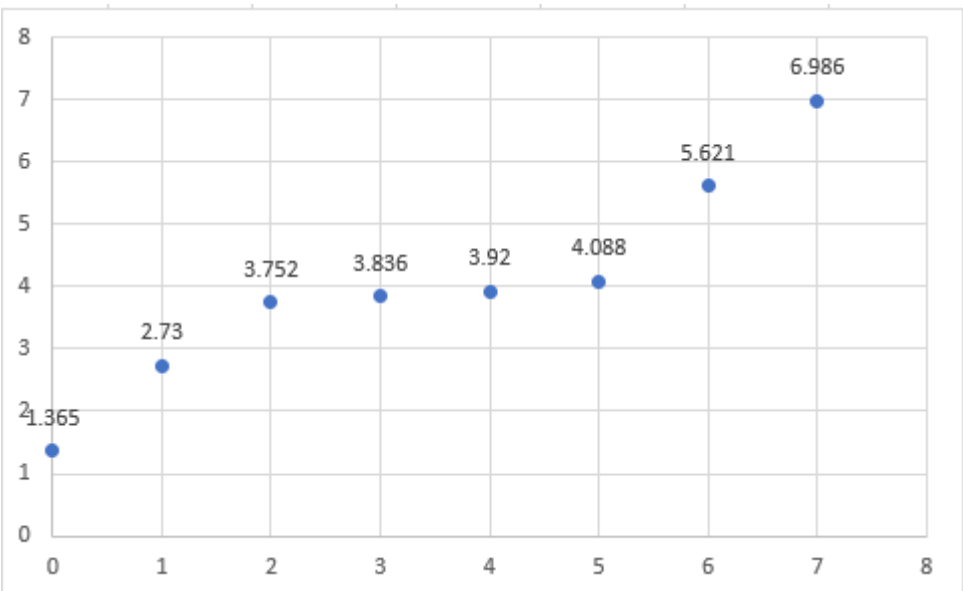
(5)

z_q	n_z	$P_z(z_q)$	$G(z_q)$	$\overline{G(z_q)}$
0	800	0.195	1.365	1
1	800	0.195	2.73	3
2	600	0.146	3.752	4
3	50	0.012	3.836	4
4	50	0.012	3.92	4
5	100	0.024	4.088	4
6	896	0.219	5.621	6
7	800	0.195	6.986	7

(PDF)



(SPV)



(6)

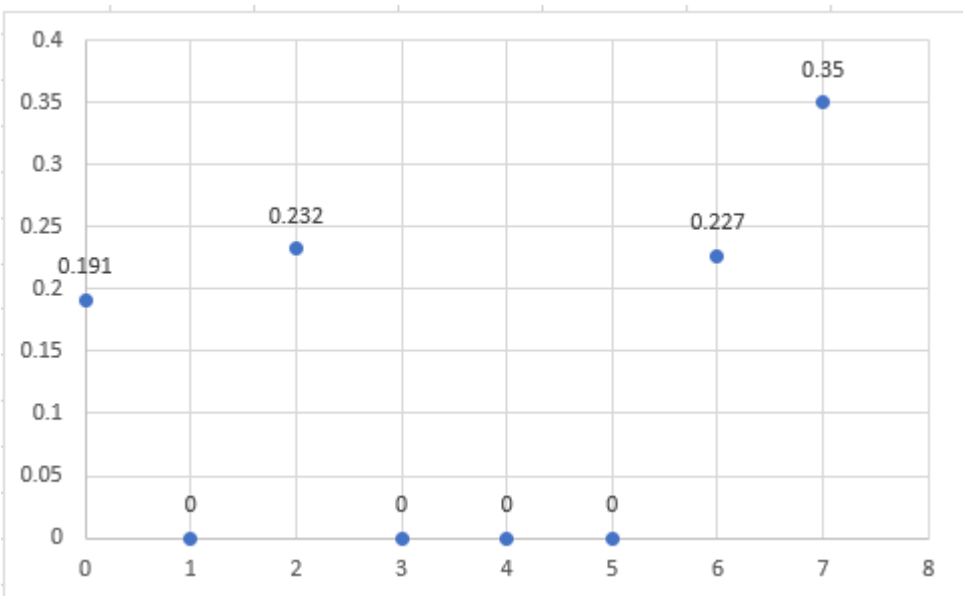
r_k	$\bar{s_k}$	$\overline{G(z_q)}$	z_q
0	1	1	0
1	3	3	1
2	5	4	2
3	6	4	3
4	7	4	4
5	7	4	5
6	7	6	6
7	7	7	7

r_k	$C_k(r_k)$
0	0.191
1	0.423
2	0.65
3	0.838
4	0.998
5	0.998
6	0.998
7	1.000

$C_z(z_q)$	z_q
0.195	0
0.39	1
0.536	2
0.548	3
0.56	4
0.584	5
0.803	6
0.998	7

r_k	z_q
0	0
1	2
2	6
3	7
4	7
5	7
6	7
7	7

z_q	$P_z(z_q)$
0	0.191
1	0
2	0.232
3	0
4	0
5	0
6	0.227
7	0.35



(III)

(1)

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt$$

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df$$

(2)

$$F(u) = \sum_{x=0}^{M-1} f(x)\, e^{-\frac{j2\pi ux}{M}}, u = 0,1,2 \dots M-1$$

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u)\, e^{\frac{j2\pi ux}{M}}, x = 0,1,2 \dots M-1$$

(3)

$\{f(0), f(1), f(2), f(3)\} = \{4,0,10,2\}$

$F(0) = 4 * e^0 + 0 * e^0 + 10 * e^0 + 2 * e^0 = 16$

$$F(1) = 4 * e^{0} + 0 * e^{-\frac{j2\pi}{4}} + 10 * e^{-\frac{j4\pi}{4}} + 2 * e^{-\frac{j6\pi}{4}}$$

$$= 4 + 0 * (-j) + 10 * (-1) + 2 * (j) = -6 + 2j$$

$$F(2) = 4 * e^{0} + 0 * e^{-\frac{j4\pi}{4}} + 10 * e^{-\frac{j8\pi}{4}} + 2 * e^{-\frac{j12\pi}{4}}$$

$$= 4 + 0 * (-1) + 10 * (1) + 2 * (-1) = 12$$

$$F(3) = 4 * e^{0} + 0 * e^{-\frac{j6\pi}{4}} + 10 * e^{-\frac{j12\pi}{4}} + 2 * e^{-\frac{j18\pi}{4}}$$

$$= 4 + 0 * (j) + 10 * (-1) + 2 * (-j) = -6 - 2j$$

$$\{F(0), F(1), F(2), F(3)\} = \{16, -6 + 2j, 12, -6 - 2j\}$$

	Multiplication	Addition
F(0)	4	3
F(1)	4	3
F(2)	4	3
F(3)	4	3
Total	16	12

(4)

$$\{f(0), f(1), f(2), f(3), f(4), f(5), f(6), f(7)\} = \{4,0,10,2,4,0,10,2\}$$

$$F(0) = 4 * e^0 + 0 * e^0 + 10 * e^0 + 2 * e^0 + 4 * e^0 + 0 * e^0 + 10 * e^0 + 2 * e^0 = 32$$

$$F(1) = 4 * e^0 + 0 * e^{-\frac{j2\pi}{8}} + 10 * e^{-\frac{j4\pi}{8}} + 2 * e^{-\frac{j6\pi}{8}} + 4 * e^{-\frac{j8\pi}{8}} + 0 * e^{-\frac{j10\pi}{8}} + 10 * e^{-\frac{j12\pi}{8}} + 2 * e^{-\frac{j14\pi}{8}}$$

$$= 4 + 0 * \frac{1}{\sqrt{2}}(1 - j) + 10 * (-j) + 2 * \frac{1}{\sqrt{2}}(-1 - j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(-1 + j) + 10 * (j) + 2 * \frac{1}{\sqrt{2}}(1 + j) = 0$$

$$F(2) = 4 * e^0 + 0 * e^{-\frac{j4\pi}{8}} + 10 * e^{-\frac{j8\pi}{8}} + 2 * e^{-\frac{j12\pi}{8}} + 4 * e^{-\frac{j16\pi}{8}} + 0 * e^{-\frac{j20\pi}{8}} + 10 * e^{-\frac{j24\pi}{8}} + 2 * e^{-\frac{j28\pi}{8}}$$

$$= [4 * (1) + 0 * (-j) + 10 * (-1) + 2 * (j)] * 2 = -12 + 4j$$

$$F(3) = 4 * e^0 + 0 * e^{-\frac{j6\pi}{8}} + 10 * e^{-\frac{j12\pi}{8}} + 2 * e^{-\frac{j18\pi}{8}} + 4 * e^{-\frac{j24\pi}{8}} + 0 * e^{-\frac{j30\pi}{8}} + 10 * e^{-\frac{j36\pi}{8}} + 2 * e^{-\frac{j42\pi}{8}}$$

$$= 4 * (1) + 0 * \frac{1}{\sqrt{2}}(-1 - j) + 10 * (j) + 2 * \frac{1}{\sqrt{2}}(1 - j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(1 + j) + 10 * (-j) + 2$$

$$* \frac{1}{\sqrt{2}}(-1 + j) = 0$$

$$F(4) = 4 * e^0 + 0 * e^{-\frac{j8\pi}{8}} + 10 * e^{-\frac{j16\pi}{8}} + 2 * e^{-\frac{j24\pi}{8}} + 4 * e^{-\frac{j32\pi}{8}} + 0 * e^{-\frac{j40\pi}{8}} + 10 * e^{-\frac{j48\pi}{8}} + 2 * e^{-\frac{j56\pi}{8}}$$

$$= [4 * (1) + 0 * (-1) + 10 * (1) + 2 * (-1)] * 2 = 24$$

$$F(5) = 4 * e^0 + 0 * e^{-\frac{j10\pi}{8}} + 10 * e^{-\frac{j20\pi}{8}} + 2 * e^{-\frac{j30\pi}{8}} + 4 * e^{-\frac{j40\pi}{8}} + 0 * e^{-\frac{j50\pi}{8}} + 10 * e^{-\frac{j60\pi}{8}} + 2 * e^{-\frac{j70\pi}{8}}$$

$$= 4 * (1) + 0 * \frac{1}{\sqrt{2}}(-1 + j) + 10 * (-j) + 2 * \frac{1}{\sqrt{2}}(1 + j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(1 - j) + 10 * (j) + 2$$

$$* \frac{1}{\sqrt{2}}(-1 - j) = 0$$

$$F(6) = 4 * e^0 + 0 * e^{-\frac{j12\pi}{8}} + 10 * e^{-\frac{j24\pi}{8}} + 2 * e^{-\frac{j36\pi}{8}} + 4 * e^{-\frac{j48\pi}{8}} + 0 * e^{-\frac{j60\pi}{8}} + 10 * e^{-\frac{j72\pi}{8}} + 2 * e^{-\frac{j84\pi}{8}}$$

$$= [4 * (1) + 0 * (j) + 10 * (-1) + 2 * (-j)] = -12 - 4j$$

$$F(7) = 4 * e^0 + 0 * e^{-\frac{j14\pi}{8}} + 10 * e^{-\frac{j28\pi}{8}} + 2 * e^{-\frac{j42\pi}{8}} + 4 * e^{-\frac{j56\pi}{8}} + 0 * e^{-\frac{j70\pi}{8}} + 10 * e^{-\frac{j84\pi}{8}} + 2 * e^{-\frac{j98\pi}{8}}$$

$$= 4 + 0 * \frac{1}{\sqrt{2}}(1 + j) + 10 * (j) + 2 * \frac{1}{\sqrt{2}}(-1 + j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(-1 - j) + 10 * (-j) + 2 * \frac{1}{\sqrt{2}}(1 - j) = 0$$

$$\{F(0), F(1), F(2), F(3), F(4), F(5), F(6), F(7)\}$$
$$= \{32, 0, -12 + 4j, 0, 24, 0, -12 - 4j, 0\}$$

(5)

$$\{f(0), f(1), f(2), f(3), f(4), f(5), f(6), f(7)\} = \{4,0,10,-2,4,0,10,-2\}$$

$$F(0) = 4 * e^0 + 0 * e^0 + 10 * e^0 - 2 * e^0 + 4 * e^0 + 0 * e^0 + 10 * e^0 - 2 * e^0 = 24$$

$$F(1) = 4 * e^0 + 0 * e^{-\frac{j2\pi}{8}} + 10 * e^{-\frac{j4\pi}{8}} - 2 * e^{-\frac{j6\pi}{8}} + 4 * e^{-\frac{j8\pi}{8}} + 0 * e^{-\frac{j10\pi}{8}} + 10 * e^{-\frac{j12\pi}{8}} - 2 * e^{-\frac{j14\pi}{8}}$$

$$= 4 + 0 * \frac{1}{\sqrt{2}}(1 - j) + 10 * (-j) - 2 * \frac{1}{\sqrt{2}}(-1 - j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(-1 + j) + 10 * (j) - 2 * \frac{1}{\sqrt{2}}(1 + j) = 0$$

$$F(2) = 4 * e^0 + 0 * e^{-\frac{j4\pi}{8}} + 10 * e^{-\frac{j8\pi}{8}} - 2 * e^{-\frac{j12\pi}{8}} + 4 * e^{-\frac{j16\pi}{8}} + 0 * e^{-\frac{j20\pi}{8}} + 10 * e^{-\frac{j24\pi}{8}} - 2 * e^{-\frac{j28\pi}{8}}$$

$$= [4 * (1) + 0 * (-j) + 10 * (-1) - 2 * (j)] * 2 = -12 - 4j$$

$$F(3) = 4 * e^0 + 0 * e^{-\frac{j6\pi}{8}} + 10 * e^{-\frac{j12\pi}{8}} - 2 * e^{-\frac{j18\pi}{8}} + 4 * e^{-\frac{j24\pi}{8}} + 0 * e^{-\frac{j30\pi}{8}} + 10 * e^{-\frac{j36\pi}{8}} - 2 * e^{-\frac{j42\pi}{8}}$$

$$= 4 * (1) + 0 * \frac{1}{\sqrt{2}}(-1 - j) + 10 * (j) - 2 * \frac{1}{\sqrt{2}}(1 - j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(1 + j) + 10 * (-j) - 2$$

$$* \frac{1}{\sqrt{2}}(-1 + j) = 0$$

$$F(4) = 4 * e^0 + 0 * e^{-\frac{j8\pi}{8}} + 10 * e^{-\frac{j16\pi}{8}} - 2 * e^{-\frac{j24\pi}{8}} + 4 * e^{-\frac{j32\pi}{8}} + 0 * e^{-\frac{j40\pi}{8}} + 10 * e^{-\frac{j48\pi}{8}} - 2 * e^{-\frac{j56\pi}{8}}$$

$$= [4 * (1) + 0 * (-1) + 10 * (1) - 2 * (-1)] * 2 = 32$$

$$F(5) = 4 * e^0 + 0 * e^{-\frac{j10\pi}{8}} + 10 * e^{-\frac{j20\pi}{8}} - 2 * e^{-\frac{j30\pi}{8}} + 4 * e^{-\frac{j40\pi}{8}} + 0 * e^{-\frac{j50\pi}{8}} + 10 * e^{-\frac{j60\pi}{8}} - 2 * e^{-\frac{j70\pi}{8}}$$

$$= 4 * (1) + 0 * \frac{1}{\sqrt{2}}(-1 + j) + 10 * (-j) - 2 * \frac{1}{\sqrt{2}}(1 + j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(1 - j) + 10 * (j) - 2$$

$$* \frac{1}{\sqrt{2}}(-1 - j) = 0$$

$$F(6) = 4 * e^0 + 0 * e^{-\frac{j12\pi}{8}} + 10 * e^{-\frac{j24\pi}{8}} - 2 * e^{-\frac{j36\pi}{8}} + 4 * e^{-\frac{j48\pi}{8}} + 0 * e^{-\frac{j60\pi}{8}} + 10 * e^{-\frac{j72\pi}{8}} - 2 * e^{-\frac{j84\pi}{8}}$$

$$= [4 * (1) + 0 * (j) + 10 * (-1) - 2 * (-j)] * 2 = -12 + 4j$$

$$F(7) = 4 * e^0 + 0 * e^{-\frac{j14\pi}{8}} + 10 * e^{-\frac{j28\pi}{8}} - 2 * e^{-\frac{j42\pi}{8}} + 4 * e^{-\frac{j56\pi}{8}} + 0 * e^{-\frac{j70\pi}{8}} + 10 * e^{-\frac{j84\pi}{8}} - 2 * e^{-\frac{j98\pi}{8}}$$

$$= 4 + 0 * \frac{1}{\sqrt{2}}(1 + j) + 10 * (j) - 2 * \frac{1}{\sqrt{2}}(-1 + j) + 4 * (-1) + 0 * \frac{1}{\sqrt{2}}(-1 - j) + 10 * (-j) - 2 * \frac{1}{\sqrt{2}}(1 - j) = 0$$

$$\{F(0), F(1), F(2), F(3), F(4), F(5), F(6), F(7)\}$$
$$= \{24, 0, -12 - 4j, 0, 32, 0, -12 + 4j, 0\}$$

(6)

Q(3) is original signal

Q(4) has double magnitude and double Δu from the original signal.

Q(5) can centralize the data.

(7)

Not even and not odd

Not even and not odd

(IV)

(1)

$$\tilde{H} = -\sum_{k=0}^{L-1} p_r(r_k)\log_2 p_r(r_k)$$

H=[0.1*$\log_2$(0.1)+0.4*$\log_2$(0.4)+0.06*$\log_2$(0.06)+0.1*$\log_2$(0.1)+0.04*$\log_2$(0.04) + 0.3*$\log_2$(0.3)]*(-1)~2.143

(2)

Coding:

Original source		Source reduction			
Symbol	Probability	1	2	3	4
a_2	0.4	0.4	0.4	0.4	0.6
a_6	0.3	0.3	0.3	0.3	0.4
a_1	0.1	0.1	0.2	0.3	
a_4	0.1	0.1	0.1		
a_3	0.06	0.1			
a_5	0.04				

Original source			Source reduction							
Symbol	Probability	Code	1		2		3		4	
a_2	0.4	1	0.4	1	0.4	1	0.4	1	0.6	0
a_6	0.3	00	0.3	00	0.3	00	0.3	00	0.4	1
a_1	0.1	011	0.1	011	0.2	010	0.3	01		
a_4	0.1	0100	0.1	0100	0.1	011				
a_3	0.06	01010	0.1	0101						
a_5	0.04	01011								

課本示範將機率較低之一端編碼為 0，亦可選擇機率較高之一端編碼為 0。但請在每一步驟以相同選擇去編碼。

課本示範每一步驟最低兩種 symbol 之機率加總後之機率若與某 symbol 之機率相同，將其擺在較低機率處；若選擇擺在較高機率處並繼續編碼，仍會給分。但請在每一步驟以相同選擇去編碼。

L_{avg} = 0.4*1+0.3*2+0.1*3+0.1*4+0.06*5+0.04*5=2.2

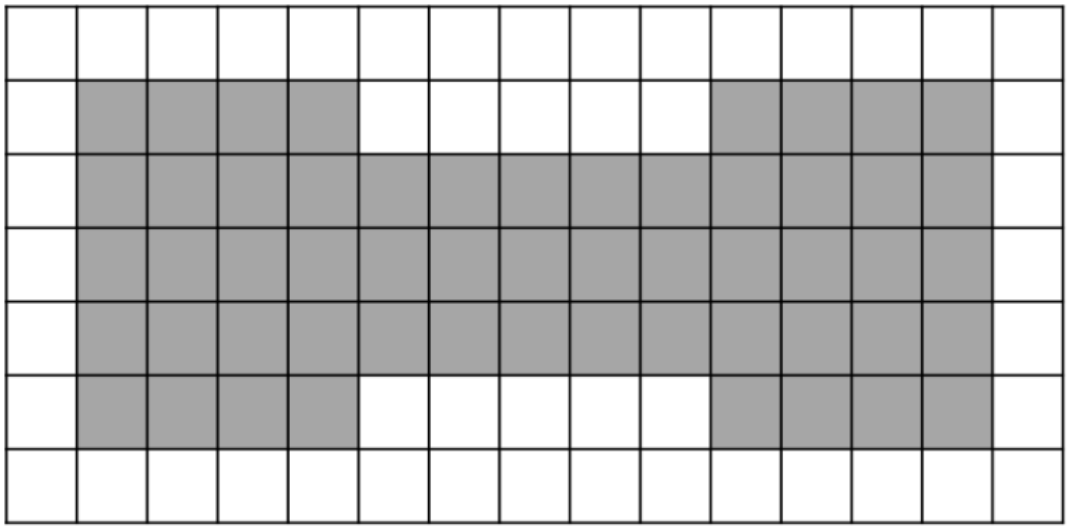
$\eta = H / L_{avg} \sim 0.974$，$\eta<1$，符合 Shannon's First Theorem

(3)

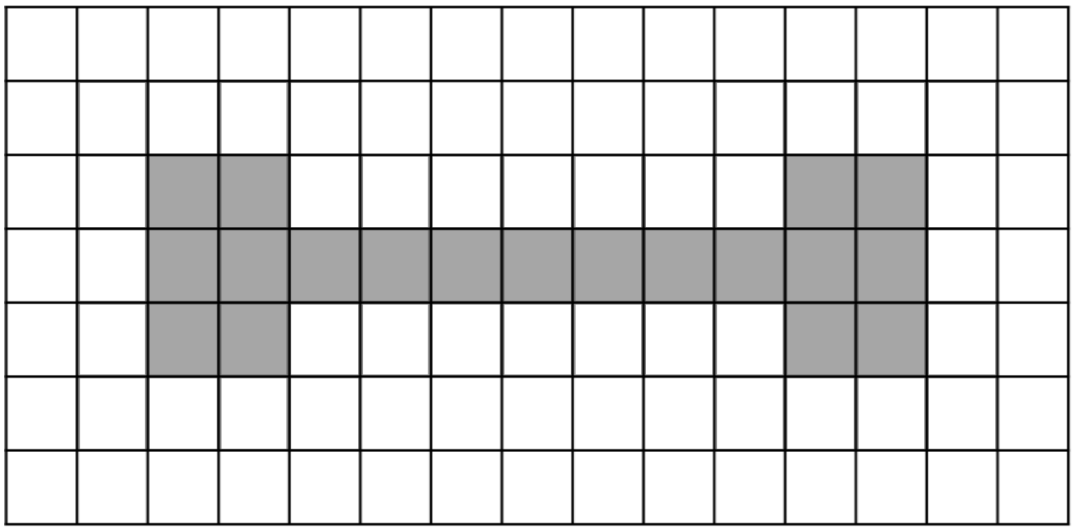
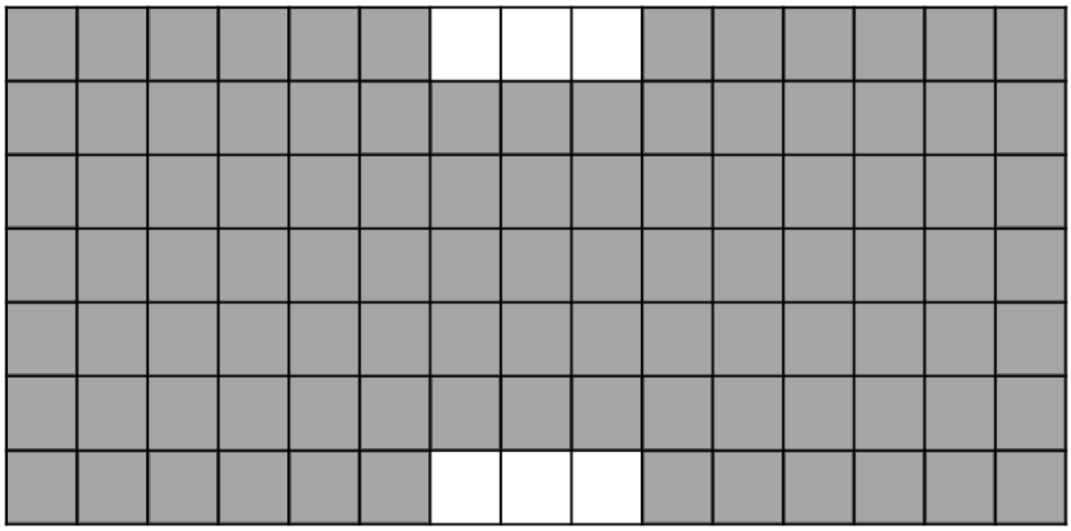
Decoded string: $a_3\ a_1\ a_4\ a_2\ a_2\ a_6$

(V)

原圖:



(1) Erosion



(2) Dilation